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Bricks Application Programming Reference

Document	Bricks Application Programming Reference
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Applies to	Bricks Transaction Server, version 1.5 and later
Companion	README.md — installation, configuration, and operations

About this document

This reference describes the application programming interface of the Bricks Transaction Server: the EXEC CICS command set, the REXX and COBOL dialects that may issue those commands, the BMS-flavoured map DSL used to build 3270 panels, and the catalogue of sample programs shipped under runtime/.

Operational and installation topics — running bricks, editing bricks.cnf, signing on through CSSN, the CEMT master terminal, the bricksload stress tester, and the /metrics endpoint — are documented in the companion README.md.

Who should read this manual

This manual is for application programmers who write transactions that run under Bricks. Familiarity with the IBM CICS programming model (pseudo-conversational dispatch, EIB, COMMAREA, BMS maps, KSDS files, temporary storage queues) is assumed; where bricks deviates from the IBM behaviour the difference is documented explicitly.

Conventions used

- **UPPERCASE** in syntax — keywords; coded literally.
- **lowercase** in syntax — supplied by the programmer (a name or expression).
- [] — optional clauses.
- { a | b } — choose one of the alternatives.
- ... — the preceding clause may repeat.
- Command syntax is shown in EXEC CICS ... END-EXEC form. The bare-string form ("VERB OPTIONS" under ADDRESS CICS, REXX only) is described in Chapter 2; both forms dispatch identically.

Notation in this manual

Each EXEC CICS command in Part 2 is documented with the same five sections, in this order:

1. **Format** — the command syntax as a code block.
 2. **Description** — what the command does and any important runtime behaviour.
 3. **Options** — every keyword that follows the verb, with its meaning and any value constraints.
 4. **Conditions** — the EIBRESP values the command may return, together with the cause of each.
 5. **Example** — a short, runnable code fragment illustrating typical use.
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- Chapter 2. The EXEC CICS command environment
- Chapter 3. The map DSL

Part 2. EXEC CICS command reference

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- Chapter 5. Program control commands — RETURN - XCTL - LINK - ABEND
- Chapter 6. System services — ASSIGN - ASKTIME - FORMATTIME
- Chapter 7. KSDS file commands — READ - WRITE - REWRITE - DELETE
- Chapter 8. KSDS browse commands — STARTBR - READNEXT - READPREV - RESETBR - ENDBR
- Chapter 9. Temporary storage commands — READQ TS - WRITEQ TS - DELETEQ TS
- Chapter 10. Recovery and condition handling — SYNCPOINT - SYNCPOINT ROLLBACK - HANDLE CONDITION - IGNORE CONDITION - HANDLE AID - HANDLE ABEND

- Chapter 11. The Execute Interface Block (EIB)
- Chapter 12. Response codes
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Part 3. The REXX language

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Part 1. The bricks programming model

Chapter 1. Overview

A bricks **transaction** is a 4-character TRANSID listed in `runtime/transactions.conf`. Each transaction names a **program** — either a REXX source file in `runtime/rexx/` or a COBOL source file in `runtime/cobol/` — and an optional access-control list:

```
HEL0:rexx:hello.rexx
HELP:rexx:help.rexx:public
QAGE:rexx:qage.rexx:public,users,admin
HELC:cobol:hello.cob:public
GUST:cobol:gust.cob:public
```

The first time a TRANSID is dispatched, its program is parsed and the AST is cached (see *Parsed-program cache* in the README); subsequent dispatches skip the parse. Each running task has its own heap and stack — there is no shared mutable state between concurrently running tasks of the same TRANSID.

Programs interact with three layers:

1. **The 3270 terminal**, through EXEC CICS SEND MAP / RECEIVE MAP (and the unstructured SEND TEXT / RECEIVE pairing). Maps are defined in the BMS-style DSL described in Chapter 3.
2. **Persistent data**, through KSDS file commands (Chapter 7) and TS-queue commands (Chapter 9). Both are backed by an embedded bbolt B+tree at data/files.boltdb.
3. **Other programs**, through LINK (synchronous, with COMMAREA), XCTL (transfer of control, no return), and RETURN TRANSID (pseudo-conversational chaining). The Execute Interface Block (EIB) exposes session and request state (Chapter 11).

REXX and COBOL share **the same EXEC CICS dispatcher** (cics/handler.go). Every command documented in Part 2 is therefore available, with identical syntax and identical semantics, from either language.

A typical pseudo-conversational task has the shape:

```
ADDRESS CICS
```

```
EXEC CICS RECEIVE MAP('CUST1')                END-EXEC    /* read prior screen */
... validate / compute ...
EXEC CICS SEND      MAP('CUST1') FROM(SCR.) ERASE END-EXEC
EXEC CICS RETURN    TRANSID('CUST')            END-EXEC    /* chain back to self */
```

The dispatcher invokes the program once per ENTER. State that must survive between invocations rides in the COMMAREA on RETURN; the chained task receives it through EIBCALEN and the DFHCOMMAREA data area.

Chapter 2. The EXEC CICS command environment

Surface forms

Bricks accepts two equivalent surface forms inside an ADDRESS CICS scope.

1. IBM-canonical EXEC CICS ... END-EXEC — the form CICS programmers recognize. Available from REXX and COBOL.

```
ADDRESS CICS
```

```
EXEC CICS ASSIGN  USERID(USR) TERMID(TRM)      END-EXEC
EXEC CICS SEND    MAP('HEL01') FROM(SCR.) ERASE END-EXEC
EXEC CICS RETURN                                     END-EXEC
```

In REXX, a small preprocessor (rexx/preprocess.go, called from rexx.Parse) rewrites each EXEC CICS ... END-EXEC block into the equivalent quoted-string command before lexing. Multi-line bodies are collapsed into a single command string; trailing newlines preserve source line numbers in error messages. Comments and string literals are left alone.

In COBOL, the parser collects every token between EXEC CICS and END-EXEC and reconstructs the body verbatim for the same cics.ParseCommand REXX uses.

2. Bare string under ADDRESS CICS — terser; available **only** in REXX.

```
ADDRESS CICS
"ASSIGN USERID(USR) TERMID(TRM)"
"SEND MAP('HEL01') FROM(SCR.) ERASE"
"RETURN"
```

Both forms route through `cics.ParseCommand` and dispatch to the same handler. After every command the handler writes `EIBRESP`, `EIBRESP2`, and `RC` into the program's frame.

Argument-passing rules

Inside the parentheses of an option:

- A bare identifier is treated as a **variable name**. The handler reads the value at runtime, and (where the option is an output) writes the result back into that variable.
- A quoted string literal ('...' or "...") is treated as a **literal value**. The handler does not write to literals; passing a literal to an output-only option is rejected with `INVREQ`.

This distinction matters most for length and key fields: `LENGTH(LEN)` both reads the requested length on input and writes the actual length back; `LENGTH(80)` only sets the input length.

Programming model summary

Concept	Bricks implementation
Task	One invocation of one <code>TRANSID</code> (<code>session.TxCB</code>).
Program	A parsed REXX or COBOL source file, cached at L1/L2.
Pseudo-conversational chain	<code>EXEC CICS RETURN TRANSID(t) COMMAREA(d)</code> .
Conversational	Loop in the program; each <code>RECEIVE MAP</code> blocks on input.
Inter-program call	<code>LINK PROGRAM(name) COMMAREA(var)</code> (synchronous).
Storage shared across tasks	KSDS files, TS queues.
Storage shared within one task	REXX variables / COBOL <code>WORKING-STORAGE</code> only.
Implicit task end	Program runs off the end, or <code>STOP RUN</code> (COBOL).
Forced task end	<code>EXEC CICS ABEND</code> .

Chapter 3. The map DSL

Bricks ships its own line-oriented, BMS-flavoured map DSL (parsed by `mapdsl/`). Maps live as `*.map` files in the directory configured by `maps_dir` (default `runtime/map/`). Each file contains one or more `MAP ... ENDMAP` blocks. Comments start with `*`.

Format

```
MAP <name> SIZE rowsxcols
  [FIELD AT r,c LEN n <attrs> "literal"]
  [INPUT <fieldname> AT r,c LEN n <attrs> [DEFAULT "value"]]
  [STOP AT r,c]
  [CURSOR AT {fieldname | r,c}]
ENDMAP
```

Statements

MAP <name> SIZE rowsxcols The map header. Names must be unique across the directory (case-insensitive); typical sizes are 24x80 (mod 2) and 43x80 (mod 4).

FIELD AT r,c LEN n <attrs> "literal" A display-only field that paints the literal at row r, column c, length n. Useful for labels, headings, and panel chrome.

INPUT <fieldname> AT r,c LEN n <attrs> [DEFAULT "value"] A named input field. The name is how SEND MAP FROM(STEM.) and RECEIVE MAP INTO(STEM.) route data to / from this position (STEM.<fieldname>). Use PROT to make the field write-protected but still named — useful for output values painted by SEND MAP FROM(STEM.).

STOP AT r,c An autoskip stop attribute that ends the preceding input field.

CURSOR AT <fieldname> or CURSOR AT r,c The home position for the cursor when the map is sent. The named form resolves to (field.Row, field.Col + 1) — i.e. one byte to the right of the leading 3270 attribute byte, which is the writable cell. Forward references are allowed; names are resolved after the whole map is parsed. When omitted, the renderer falls back to the first input field.

ENDMAP Terminates the map.

Attributes

PROT, UNPROT, BRIGHT, DIM, UNDERSCORE, HIDDEN, NUMERIC, MDT, BLINK, REVERSE, COLOR=BLUE|RED|PINK|GREEN|TURQUOISE|YELLOW|WHITE.

Catalogue lifecycle

mapdsl.NewCatalog(dir) parses every *.map file at startup and self-refreshes on subsequent edits. Each Lookup(name) stats the directory plus the source file backing the requested name; on mtime change the directory is reparsed and swapped atomically. A failed re-parse keeps the prior catalogue in place — the operator can find the broken file with CEMT PERFORM RESCAN MAP (see the README).

Example

```
* CICS sign-on screen
MAP CSSN SIZE 24x80
  FIELD AT 1,28 LEN 24 PROT BRIGHT COLOR=TURQUOISE "BRICKS SIGN-ON"
  FIELD AT 6,15 LEN 9  PROT "Userid:"
```



```

INPUT USERID    AT 6,25 LEN 8 UNDERSCORE COLOR=GREEN
STOP            AT 6,34
FIELD AT 8,15 LEN 9 PROT                                "Password:"
INPUT PASSWORD  AT 8,25 LEN 8 HIDDEN UNDERSCORE COLOR=RED
STOP            AT 8,34
CURSOR          AT USERID
ENDMAP

```

Part 2. EXEC CICS command reference

This part documents every EXEC CICS command bricks implements. Each command page follows the same layout: **Format**, **Description**, **Options**, **Conditions**, **Example**.

Commands are grouped by function:

- Chapter 4. Terminal I/O
- Chapter 5. Program control
- Chapter 6. System services
- Chapter 7. KSDS files
- Chapter 8. KSDS browse
- Chapter 9. Temporary storage

Cross-cutting reference:

- Chapter 10. Recovery and condition handling
 - Chapter 11. The EIB
 - Chapter 12. Response codes
 - Chapter 13. Commands not implemented
-

Chapter 4. Terminal I/O commands

SEND MAP

Send a BMS-style map to the 3270 terminal and wait for the operator to press an AID key.

Format

```

EXEC CICS SEND MAP(name)
                    [FROM(stem.)]
                    [ERASE]
                    [CURSOR(position)]
END-EXEC

```

Description Loads the named map from the catalogue, populates each named field with the value of <stem>.<fieldname> (when FROM is supplied), paints the screen, and waits for the operator to press an AID key (ENTER, PFn, PAn, CLEAR). On return, the captured response is

stored on the TCB so a subsequent RECEIVE MAP can pull modified field values back into the program. EIBAID and EIBCP0SN are updated with the AID character and 1-based cursor position.

If FROM is omitted, the map renders using its INPUT DEFAULT values. If FROM is a literal string, every named field on the map is filled with that literal (rare; supported for parity with BMS).

ERASE clears the screen first; without ERASE existing fields stay behind underneath the new map (3270 NoClear).

Options **MAP(name)** — *required* Name of a map in the catalogue (case-insensitive). MAPFAIL if absent.

FROM(stem.) Stem variable whose tails name the map's input fields. The stem's trailing dot is optional: FROM(SCR) and FROM(SCR.) are equivalent.

ERASE Clears the screen before painting.

CURSOR(position) 1-based cursor position. Overrides the map's CURSOR AT clause.

Conditions

Condition	EIBRESP	Cause
NORMAL	0	Map sent, AID received.
MAPFAIL	36	Named map not found in the catalogue.
IOERR	17	The terminal write failed.

Example

```
SCR.GREETING = 'Hello, ' || USR  
EXEC CICS SEND MAP('HEL01') FROM(SCR.) ERASE END-EXEC
```

RECEIVE MAP

Retrieve the operator's input from the most recent SEND MAP.

Format

```
EXEC CICS RECEIVE MAP(name)  
                [INTO(stem.)]  
END-EXEC
```

Description Pulls the response stored by the most recent SEND MAP into a stem, one tail per named field on the map (<stem>.<fieldname>). When INTO is omitted, the default stem is MAP..

A RECEIVE MAP with no matching prior SEND MAP returns MAPFAIL.

Options **MAP(name)** — *required* The same map name passed to the prior SEND MAP.
INTO(stem.) Destination stem. Default MAP..

Conditions

Condition	EIBRESP	Cause
NORMAL	0	Fields retrieved.
MAPFAIL	36	No prior matching SEND MAP, or named map not found.

Example

```
EXEC CICS RECEIVE MAP('CUST1') INTO(IN.) END-EXEC
ACTION = IN.ACTION
CKEY   = IN.CUSTNO
```

SEND TEXT

Free-form text output without a BMS map.

Format

```
EXEC CICS SEND TEXT FROM(area)
                        [LENGTH(n)]
                        [ERASE]
END-EXEC
```

Description Real-CICS SEND TEXT. The body is treated as a flat row-major buffer: every cols bytes (default 80) lands on the next row, starting at row 0. **3270 has no LF/CR**, so programs must pad each logical line to the column width and concatenate (REXX LEFT(s,80), COBOL PIC X(80) group children).

LENGTH(n) truncates the body to n bytes; bytes past rows*cols are dropped. ERASE clears the screen before painting; without it, existing fields stay behind. Like SEND MAP, the call paints synchronously and waits for an AID, so the operator has time to read the screen before RETURN.

Works identically from REXX and COBOL.

Options **FROM(area)** — *required* The buffer to paint. Can be a REXX variable or a COBOL group item.

LENGTH(n) Truncate the body to this length.

ERASE Clear the screen first.

Conditions

Condition	EIBRESP	Cause
NORMAL	0	Body painted, AID received.
INVREQ	16	FROM missing.
IOERR	17	The terminal write failed.

Example See Chapter 27. Worked examples, the GETC program, for the canonical multi-row SEND TEXT pattern.

RECEIVE

Retrieve the unedited terminal line typed by the operator at the blank prompt — typically used to pick up command-line arguments.

Format

```
EXEC CICS RECEIVE INTO(buffer)
                        [LENGTH(len)]
END-EXEC
```

Description Returns the unedited terminal line the operator typed at the blank prompt (TRANSID prefix included), e.g. EXAM 1 2 3. Single-shot per task: a second RECEIVE in the same task, or any RECEIVE in a chained RETURN TRANSID task, returns EOC (RESP=6).

When LENGTH(var) is a bare variable, the actual byte count of the line is written back. The receiving COBOL field's PIC X(n) width handles padding/truncation via standard MOVE semantics; programs detect truncation by comparing len against n.

Options INTO(buffer) — *required* Destination variable for the line.

LENGTH(len) When a bare variable, receives the actual length on return.

Conditions

Condition	EIBRESP	Cause
NORMAL	0	Line returned.
EOC	6	No input available (already consumed in this task chain).

Example

```
EXEC CICS RECEIVE INTO(BUF) LENGTH(LEN) END-EXEC /* "GETC 100" */
PARSE VAR BUF TID CKEY .
```

```
EXEC CICS RECEIVE INTO(WS-INPUT) LENGTH(WS-LEN) END-EXEC
UNSTRING WS-INPUT DELIMITED BY ' '
      INTO WS-TID WS-A WS-B WS-C
END-UNSTRING.
```

Chapter 5. Program control commands

RETURN

End the current task; optionally chain to another TRANSID with a COMMAREA.

Format

```
EXEC CICS RETURN [TRANSID(id)]
                  [COMMAREA(data)]
END-EXEC
```

Description Sets `tcb.NextTransid` and `tcb.Commarea` and ends the task. With no TRANSID, control falls back to the blank prompt. With a TRANSID, the dispatcher invokes that TRANSID next, with the supplied COMMAREA available to the new task as DFHCOMMAREA and its length as EIBCALEN.

Options **TRANSID(id)** The next TRANSID to dispatch.

COMMAREA(data) Bytes to flow into the next task.

Conditions

Condition	EIBRESP	Cause
NORMAL	0	Task ended.

Example

```
EXEC CICS RETURN TRANSID('CUST') COMMAREA(SAVEAREA) END-EXEC
```

XCTL

Transfer control to another program. The current program does not resume.

Format

```
EXEC CICS XCTL PROGRAM(name)
                  [COMMAREA(data)]
END-EXEC
```

Description Transfer of control. The named program runs in place of the current one; the COMMAREA, if supplied, becomes its DFHCOMMAREA. The current task's chain continues with the new program; RETURN TRANSID semantics carry through unchanged.

Options **PROGRAM(name)** — *required* The target program. Resolved through runtime/transactions.conf.

COMMAREA(data) Bytes to flow to the target.

Conditions

Condition	EIBRESP	Cause
NORMAL	0	Control transferred.
PGMIDERR	27	Target program not in transactions.conf.
INVREQ	16	PROGRAM missing.

Example

```
EXEC CICS XCTL PROGRAM('VALIDATE') COMMAREA(SCR) END-EXEC
```

LINK

Synchronously call a sub-program with bidirectional COMMAREA.

Format

```
EXEC CICS LINK PROGRAM(name)
                [COMMAREA(var | 'literal')]
END-EXEC
```

Description The target program runs in a fresh frame with its DFHCOMMAREA preloaded from the caller's COMMAREA(var) (or literal). When the sub-program exits, its final DFHCOMMAREA is written back to the caller's variable.

Caller state — NextTransid, Commarea, LastResponse, LastMapName — is saved before the LINK and restored on return, so the LINK is transparent to the caller's pseudo-conversational context.

The per-transaction ACL is rechecked on the target so a low-privilege caller cannot escalate by linking into an admin program.

REXX and COBOL programs can LINK to each other freely; DFHCOMMAREA is marshalled as opaque bytes.

Options **PROGRAM(name)** — *required* The sub-program to call. Resolved through run-time/transactions.conf.

COMMAREA(var | 'literal') Bidirectional data area. A bare variable is read on entry and written on exit; a literal is read-only.

Conditions

Condition	EIBRESP	Cause
NORMAL	0	Sub-program returned cleanly.
PGMIDERR	27	Target not in transactions.conf, sub-program errored, or the caller is not authorised for the target.

Example

```
SAV = ''                                /* working area */
EXEC CICS LINK PROGRAM('CUSV') COMMAREA(SAV) END-EXEC
IF EIBRESP <> 0 THEN SIGNAL ERR
PARSE VAR SAV STATUS '|' MSG
```

ABEND

Abnormally terminate the current task.

Format

```
EXEC CICS ABEND [ABCODE(code)]
                [NODUMP]
END-EXEC
```

Description Ends the task with the supplied 4-character abend code (default AAAA). Bricks logs the abend on the operator console and reflects it in the TxCB's status; pseudo-conversational chains are broken (no NextTransid is honoured after an ABEND).

Options **ABCODE(code)** 4-character code to record. Defaults to AAAA.

NODUMP Accepted for parity with IBM CICS; no dumps are produced anyway.

Conditions ABEND does not return; nothing follows it in the program flow.

Example

```
IF EIBRESP <> 0 THEN  
  EXEC CICS ABEND ABCODE('FI01') END-EXEC
```

Chapter 6. System services

ASSIGN

Read EIB, session, and environment fields into the program.

Format

```
EXEC CICS ASSIGN <FIELD>(target) [<FIELD>(target) ...]  
END-EXEC
```

Description Reads one or more system fields into program variables. Multiple options can be combined in a single ASSIGN call. Each <FIELD> is one of the keywords below; target is the variable that receives the value.

Options

Option	Returns
USERID(t)	Signed-on userid (empty until CSSN succeeds).
TERMID(t) / EIBTRMID(t) EIBAID(t)	Unique 4-digit terminal id (T0001 ...). Single-byte AID character of the most recent SEND MAP / RECEIVE MAP. Compare with C2X(EIBAID) = 'F3' (REXX) or EIBAID = X'F3' (COBOL) to detect PF3, etc.
EIBCP0SN(t)	1-based cursor position from the most recent map response.
EIBCALEN(t)	Length of DFHCOMMAREA flowed in from the caller.
TWALENG(t) / TCTUALENG(t)	Always 0 (bricks does not allocate a TWA / TCTUA).
SCREENHT(t) / SCREENWD(t) ALTSCRNHT(t) / ALTSCRNWD(t)	Negotiated terminal rows / columns. Same values; bricks does not distinguish primary and alternate sizes.
CONNECTED(t) / CONNTIME(t)	Wall-clock connect timestamp YYYY-MM-DD HH:MM:SS.
TLS(t)	yes when the session is on the TLS listener, no otherwise.
DATE(t)	Today as YYYYMMDD. (<i>bricks-specific</i>)
TIME(t)	Now as HHMMSS. (<i>bricks-specific</i>)

Option	Returns
TODAYYR(t) / TODAYMO(t) / TODAYDY(t)	Today's year / month / day individually. (bricks-specific)
DAYCOUNT(t)	Days since 1970-01-01. Subtract two values for an exact day delta. (bricks-specific)

The bricks-specific options exist primarily so the COBOL subset can do date math without REXX-style intrinsic functions; see `runtime/cobol/qagc.cob` for an example that computes age in years from a birthdate.

Conditions

Condition	EIBRESP	Cause
NORMAL	0	All requested fields returned.
INVREQ	16	A field name is unknown, or its target is a literal rather than a variable.

Example

```
EXEC CICS ASSIGN USERID(USR)
                TERMID(TRM)
                SCREENHT(SCRH)
                SCREENWD(SCRW)
END-EXEC
```

ASKTIME

Refresh the EIB date/time fields and, optionally, return the current time as a 15-digit absolute timestamp (ABSTIME).

Format

```
EXEC CICS ASKTIME [ABSTIME(target)]
END-EXEC
```

Description ASKTIME re-reads the system clock and writes the current date and time into EIB-DATE and EIBTIME. Both fields follow the IBM CICS formats:

- EIBDATE — packed-style decimal 0CYYYYY, where C is (century - 19) and YYYY is the two-digit year and Julian day-of-year. For 2026-05-12 (Julian day 132), EIBDATE = 1026132.
- EIBTIME — six-digit HHMMSS.

If `ABSTIME(target)` is given, `target` receives a 15-digit decimal string representing milliseconds since 1900-01-01 00:00:00.000. Pass that value to `FORMATTIME` to break it back into formatted date / time components.

`ASKTIME` is the only way to get a fresh `ABSTIME`; the bricks-specific `ASSIGN DATE / TIME` shortcuts return formatted strings only.

Options **`ABSTIME(target)`** Variable that receives the 15-character absolute time. Optional; when omitted, only `EIBDATE` and `EIBTIME` are refreshed.

Conditions

Condition	EIBRESP	Cause
NORMAL	0	Always (clock reads cannot fail).

Example

```
EXEC CICS ASKTIME ABSTIME(NOW) END-EXEC
EXEC CICS FORMATTIME ABSTIME(NOW)
                        YYYYMMDD(TODAY)
                        TIME(CLOCK) TIMESEP(':')
END-EXEC
SAY 'Stamped at' TODAY CLOCK
```

FORMATTIME

Decode an `ABSTIME` value into formatted date and time fields.

Format

```
EXEC CICS FORMATTIME ABSTIME(source)
                        [DATE(t)] [DATEFORM(fmt)] [DATESEP(c)]
                        [YYYYMMDD(t)] [MMDDYYYY(t)] [DDMMYYYY(t)]
                        [MMDDYY(t)] [DDMMYY(t)]
                        [YYYYDDD(t)] [YYDDD(t)]
                        [TIME(t)] [TIMESEP(c)]
                        [YEAR(t)] [MONTHOFYEAR(t)] [DAYOFMONTH(t)]
                        [DAYOFWEEK(t)] [DAYCOUNT(t)]
END-EXEC
```

Description `FORMATTIME` is the IBM-standard companion to `ASKTIME`. Given a 15-digit `ABSTIME` value (typically returned by `ASKTIME ABSTIME(...)`, but any 15-digit decimal millisecond stamp is accepted), it writes the requested representations into the named target variables.

Any combination of output options may be requested in a single call; options that are omitted cost nothing.

Options **ABSTIME(source)** (*required*) The 15-digit decimal absolute time to decode.

DATE(t) with optional **DATEFORM(fmt)** and **DATESEP(c)** Generic date target. DATEFORM is one of YYYYMMDD (default), MMDDYYYY, DDMMYYYY, MMDDYY, DDMMYY. DATESEP (c) inserts a one-character separator between components (e.g. '-' □ 2026-05-12); omit DATESEP for a packed string with no separators.

YYYYMMDD / MMDDYYYY / DDMMYYYY / MMDDYY / DDMMYY Direct date format targets; each respects DATESEP (c) if given.

YYYYDDD(t) / YYDDD(t) Julian-style date with three-digit day-of-year. Honour DATESEP.

TIME(t) with optional **TIMESEP(c)** Six-digit HHMMSS, or with TIMESEP(':') formatted as HH:MM:SS.

YEAR(t) — four-digit year (2026). **MONTHOFYEAR(t)** — 1..12. **DAYOFMONTH(t)** — 1..31. **DAYOFWEEK(t)** — 0 (Sunday) .. 6 (Saturday), per IBM CICS. **DAYCOUNT(t)** — days since 1900-01-01 (signed, integer).

Conditions

Condition	EIBRESP	Cause
NORMAL	0	All requested fields written.
INVREQ	16	ABSTIME is missing, not 15 decimal digits, or out of range.

Example

```
EXEC CICS ASKTIME ABSTIME(WS-NOW) END-EXEC
EXEC CICS FORMATTIME ABSTIME(WS-NOW)
                        DATE(WS-DATE) DATEFORM('DDMMYYYY') DATESEP('/')
                        TIME(WS-TIME) TIMESEP(':')
                        DAYOFWEEK(WS-DOW)
END-EXEC.
* WS-DATE → "12/05/2026"      WS-TIME → "10:30:45"      WS-DOW → "2"
```

Chapter 7. KSDS file commands

These commands operate on a **single record**, identified by a key, in a CICS FILE. Each FILE is a bbolt bucket inside data/files.boltdb; record bodies are opaque bytes (the application chooses the layout). See *How file storage works* in the README for the on-disk model.

READ

Read a record from a KSDS by key.

Format

```
EXEC CICS READ FILE(name)
           {INTO(target) | SET(target)}
           RIDFLD(key)
           [UPDATE]
           [LENGTH(len)]
END-EXEC
```

Description B+tree exact-key lookup, $O(\log n)$. The record bytes (whatever the application stored) come back into the target. UPDATE records a per-session lock on the key, gating a subsequent REWRITE on the same FILE.

When LENGTH(var) is a bare variable, the actual record length is written back.

Options **FILE(name)** — *required* The CICS FILE name.

INTO(target) / SET(target) — *one required* Destination for the record body. INTO and SET are accepted interchangeably here.

RIDFLD(key) — *required* The key to look up.

UPDATE Record a per-session lock so a subsequent REWRITE can update this record.

LENGTH(len) Receives the record's actual length when the option is a bare variable.

Conditions

Condition	EIBRESP	Cause
NORMAL	0	Record returned.
NOTFND	13	Key not present.
INVREQ	16	Missing required option, or invalid file name.
IOERR	17	Underlying store error.

Example

```
EXEC CICS READ FILE('CUSTOMERS')
           INTO(REC) RIDFLD(CKEY)
END-EXEC
IF EIBRESP = 13 THEN MSG = 'Customer not found'
```

WRITE

Insert a new record into a KSDS.

Format

```
EXEC CICS WRITE FILE(name)
                FROM(data)
                RIDFLD(key)
END-EXEC
```

Description B+tree insert. The bucket for the FILE is created implicitly on first WRITE — there is no EXEC CICS DEFINE FILE step. DUPREC if a record with the same key already exists.

The write commits in a single bbolt Update transaction with fsync on success.

Options **FILE(name)** — *required* Target FILE. Created if absent.

FROM(data) — *required* The record bytes to store.

RIDFLD(key) — *required* The key under which to store the record.

Conditions

Condition	EIBRESP	Cause
NORMAL	0	Record written.
DUPREC	14	A record with this key already exists.
INVREQ	16	Missing required option, or invalid name.
IOERR	17	Underlying store error.

Example

```
REC = NM || '|' || AD || '|' || CY || '|' || PH
EXEC CICS WRITE FILE('CUSTOMERS') FROM(REC) RIDFLD(CKEY) END-EXEC
```

REWRITE

Replace the record locked by the most recent READ ... UPDATE.

Format

```
EXEC CICS REWRITE FILE(name)
                FROM(data)
END-EXEC
```

Description Overwrites the value at the key locked by the most recent READ FILE ... UPDATE on the same FILE. Releases the per-FCB update lock at end of transaction.

Options **FILE(name)** — *required* **FROM(data)** — *required*

Conditions

Condition	EIBRESP	Cause
NORMAL	0	Record updated.
INVREQ	16	No prior READ ... UPDATE, or the lock has been released.
IOERR	17	Underlying store error.

Example

```
EXEC CICS READ FILE('CUSTOMERS') INTO(REC) RIDFLD(CKEY) UPDATE END-EXEC
PARSE VAR REC NM '|' AD '|' CY '|' PH
PH = NEWPHONE                                /* mutate */
REC = NM || '|' || AD || '|' || CY || '|' || PH
EXEC CICS REWRITE FILE('CUSTOMERS') FROM(REC) END-EXEC
```

DELETE

Remove a record from a KSDS.

Format

```
EXEC CICS DELETE FILE(name)
                        [RIDFLD(key)]
END-EXEC
```

Description If RIDFLD is supplied, deletes that key. Otherwise deletes the key locked by the most recent READ ... UPDATE on the same FILE.

Options **FILE(name)** — *required*

RIDFLD(key) Optional. When omitted, the most recent READ-UPDATE key is used.

Conditions

Condition	EIBRESP	Cause
NORMAL	0	Record deleted.
NOTFND	13	Key not present.
INVREQ	16	No RIDFLD and no prior READ-UPDATE; or invalid name.
IOERR	17	Underlying store error.

Example

```
EXEC CICS DELETE FILE('CUSTOMERS') RIDFLD(CKEY) END-EXEC
```

Chapter 8. KSDS browse commands

A **browse** walks a CICS FILE in B+tree key order. The browse runs inside a bbolt MVCC read transaction, so the cursor sees a stable point-in-time snapshot — concurrent WRITE / REWRITE / DELETE on the same FILE do not disturb an in-progress browse.

The browse is **per-task** and tied to one FILE. A program may have multiple browses open on different FILES at once; a second STARTBR on the same FILE replaces the first (STARTBR is implicitly idempotent in CICS).

The dispatcher releases any cursor the program forgot to ENDBR via a defer handler. `CloseBrowses()` at task end.

STARTBR

Open a browse cursor on a KSDS file.

Format

```
EXEC CICS STARTBR FILE(name)
                        [RIDFLD(start)]
                        [GTEQ | EQUAL]
                        [GENERIC]
                        [KEYLENGTH(n)]
END-EXEC
```

Description Opens a B+tree browse cursor on the file. With no RIDFLD, positions on the first key. With RIDFLD and GTEQ (the IBM default and bricks default), positions on the first key $\geq$ start. With EQUAL, requires an exact match (NOTFND if absent). With GENERIC and KEYLENGTH(n), positions on (and walks only through) keys whose first n bytes match the first n bytes of RIDFLD.

Options **FILE(name)** — *required*

RIDFLD(start) Starting key. Default: first key in the file.

GTEQ | EQUAL Comparison rule. Default GTEQ.

GENERIC Restrict the walk to keys whose prefix matches.

KEYLENGTH(n) With GENERIC, the prefix length to match.

Conditions

Condition	EIBRESP	Cause
NORMAL	0	Cursor opened.
NOTFND	13	EQUAL requested but no such key.
INVREQ	16	Missing required option, invalid name, etc.
IOERR	17	Underlying store error.

Example

```
EXEC CICS STARTBR FILE('CUSTOMERS')
      RIDFLD('NY-')
      GENERIC KEYLENGTH(3)
END-EXEC
```

READNEXT

Step the open browse cursor forward by one record.

Format

```
EXEC CICS READNEXT FILE(name)
      {INTO(target) | SET(target)}
      [RIDFLD(keyvar)]
      [LENGTH(len)]
END-EXEC
```

Description Advances the cursor and returns the next record (or the first record if this is the first READNEXT after STARTBR). When RIDFLD is a bare variable, the matching key is written back; when LENGTH is a bare variable, the actual record length is written back.

Returns ENDFILE past the last key (or past the end of the GENERIC prefix). Records that were deleted by a concurrent transaction between STARTBR and the read are skipped automatically (bounded forward loop, no goroutine-stack risk).

Options **FILE(name)** — *required*

INTO(target) / SET(target) — *one required*

RIDFLD(keyvar) Receives the key of the record returned.

LENGTH(len) Receives the actual record length.

Conditions

Condition	EIBRESP	Cause
NORMAL	0	Record returned.
ENDFILE	20	Past the last key (or out of GENERIC prefix).

Condition	EIBRESP	Cause
INVREQ	16	No STARTBR is open on this FILE.
IOERR	17	Underlying store error.

Example See STARTBR example, then:

```
DO FOREVER
  EXEC CICS READNEXT FILE('CUSTOMERS') INTO(REC) RIDFLD(K) END-EXEC
  IF EIBRESP = 20 THEN LEAVE
  SAY K ':' REC
END
```

READPREV

Step the open browse cursor backward by one record.

Format

```
EXEC CICS READPREV FILE(name)
                        {INTO(target) | SET(target)}
                        [RIDFLD(keyvar)]
                        [LENGTH(len)]
END-EXEC
```

Description Same writeback rules as READNEXT. Returns ENDFILE before the first key (or before the start of the GENERIC prefix). Useful for paginating backward through a key range.

Options As for READNEXT.

Conditions As for READNEXT, with ENDFILE meaning *before the first key*.

Example

```
EXEC CICS READPREV FILE('CUSTOMERS') INTO(REC) RIDFLD(K) END-EXEC
```

RESETBR

Reposition the cursor of an open browse without closing the underlying read transaction.

Format

```
EXEC CICS RESETBR FILE(name)
      RIDFLD(start)
      [GTEQ | EQUAL]
      [GENERIC]
      [KEYLENGTH(n)]
END-EXEC
```

Description Cheaper than ENDBR + STARTBR when the program wants to jump within the same browse session — the read snapshot is preserved.

Options As for STARTBR. RIDFLD is required.

Conditions

Condition	EIBRESP	Cause
NORMAL	0	Cursor repositioned.
INVREQ	16	No STARTBR is open on this FILE.
NOTFND	13	EQUAL requested but no such key.

Example

```
EXEC CICS RESETBR FILE('CUSTOMERS') RIDFLD('00500') END-EXEC
```

ENDBR

Close an open browse cursor.

Format

```
EXEC CICS ENDBR FILE(name)
END-EXEC
```

Description Releases the cursor and the underlying bbolt read transaction. The dispatcher releases any cursor the program forgot to ENDBR at task end, but explicit ENDBR is recommended.

Options **FILE(name)** — *required*

Conditions

Condition	EIBRESP	Cause
NORMAL	0	Cursor closed.

Condition	EIBRESP	Cause
INVREQ	16	No STARTBR is open on this FILE.

Example

```
EXEC CICS ENDBR FILE('CUSTOMERS') END-EXEC
```

Chapter 9. Temporary storage commands

Temporary Storage (TS) queues are ordered, persistent sequences of opaque byte items. Each queue is a bbolt sub-bucket whose keys are 8-byte big-endian item numbers (1, 2, 3 ...) and whose values are the item payloads. A single queue scales to millions of items without filesystem-directory pressure; reads and writes are $O(\log N)$.

Bricks implements only the **TS** form. The **TD** (transient data) sub-form is parsed but explicitly rejected with INVREQ — see Chapter 13. Commands not implemented.

READQ TS

Read an item from a temporary storage queue.

Format

```
EXEC CICS READQ TS QUEUE(name)
                        INTO(target)
                        [ITEM(n) | NEXT]
                        [LENGTH(len)]
                        [NUMITEMS(num)]
END-EXEC
```

QNAME is accepted as a synonym for QUEUE.

Description Reads one item from the queue. With ITEM(n), returns item n. Without ITEM, or with NEXT, advances the **per-task implicit cursor**: the first cursor-less READQ on a queue returns item 1, the second returns item 2, and so on. The cursor is keyed on the running TxCB and released when the task ends — a fresh invocation of the same TRANSID starts at item 1 again.

Options **QUEUE(name)** / **QNAME(name)** — *required*

INTO(target) — *required*

ITEM(n) | **NEXT** Item to read. Default = next per the implicit cursor.

LENGTH(len) Receives the actual item length.

NUMITEMS(num) Receives the queue's current high-water item count.

Conditions

Condition	EIBRESP	Cause
NORMAL	0	Item returned.
ITEMERR	26	ITEM(n) out of range, or implicit cursor past the last item.
QIDERR	44	Queue does not exist.
INVREQ	16	Missing required option, invalid name.

Example

```
DO FOREVER
  EXEC CICS READQ TS QUEUE(QNM) INTO(REC) END-EXEC
  IF EIBRESP = 26 THEN LEAVE          /* end of queue */
  SAY REC
END
```

WRITEQ TS

Append a new item to a queue, or rewrite an existing item.

Format

```
EXEC CICS WRITEQ TS QUEUE(name)
                        FROM(data)
                        [ITEM(n) REWRITE]
END-EXEC
```

QNAME is accepted as a synonym for QUEUE.

Description In **append** mode (no ITEM REWRITE), the item is added at the next sequence number; an in-memory high-water-mark counter assigns the number without scanning. When ITEM(var) is a bare variable, the assigned item number is written back.

In **rewrite** mode (both ITEM(n) and REWRITE present), item n is replaced.

The write commits in a single bbolt transaction so a crash mid-write leaves either the prior state or the new one — never a partial.

Options QUEUE(name) / QNAME(name) — *required* FROM(data) — *required* ITEM(n) REWRITE Both required for rewrite mode; either alone is rejected.

Conditions

Condition	EIBRESP	Cause
NORMAL	0	Item written.
ITEMERR	26	ITEM(n) REWRITE and item n does not exist.
INVREQ	16	Missing required option, invalid name, or ITEM/REWRITE not paired.
IOERR	17	Underlying store error.

Example

```
EXEC CICS WRITEQ TS QUEUE('AUDIT') FROM(MSG) ITEM(I) END-EXEC
SAY 'Wrote item' I
```

DELETEQ TS

Delete an entire temporary storage queue.

Format

```
EXEC CICS DELETEQ TS QUEUE(name) END-EXEC
```

Description Drops the queue's sub-bucket and resets the in-memory counters and cursors. Subsequent WRITEQ will recreate it starting at item 1.

Options **QUEUE(name)** / **QNAME(name)** — *required*

Conditions

Condition	EIBRESP	Cause
NORMAL	0	Queue deleted.
QIDERR	44	Queue does not exist.

Example

```
EXEC CICS DELETEQ TS QUEUE('AUDIT') END-EXEC
```

Chapter 10. Recovery and condition handling

This chapter covers two related families:

- **Unit-of-work commands** — SYNCPOINT and SYNCPOINT ROLLBACK group multiple file / TS mutations into one atomic step that can be committed or undone.

- **Non-local control commands** — `HANDLE CONDITION`, `IGNORE CONDITION`, `HANDLE AID`, and `HANDLE ABEND` arm program labels that the dispatcher branches to on a non-NORMAL response, on a particular AID key, or when the task abends. Together with the per-call `EIBRESP` test (Chapter 12) and `REXX SIGNAL ON ERROR` (Chapter 18) they cover the full range of CICS error-handling idioms.

How the unit of work works

Each task owns a **journal** of pending undo entries that lives on its TxCB. Every successful `WRITE` / `REWRITE` / `DELETE` and every `WRITEQ TS` / `DELETEQ TS` appends an inverse operation to the journal *immediately after* the underlying bbolt write commits.

- `SYNCPPOINT` clears the journal — the work becomes irrevocably committed, and a new unit of work begins.
- `SYNCPPOINT ROLLBACK` walks the journal in reverse and applies each inverse op (re-writing pre-images, re-creating deleted records, restoring queue contents, etc.), then clears the journal.
- `RETURN` performs an **implicit SYNCPPOINT** before the task ends.
- `ABEND` performs an **implicit SYNCPPOINT ROLLBACK** *unless* a `HANDLE ABEND` exit catches it; the exit may then choose whether to commit or roll back explicitly.

In-flight uncommitted writes are visible to concurrent tasks (this matches IBM CICS under `READ NO UPDATE`); the rollback model exists to recover *the current task* from its own partially-applied work, not to provide multi-task isolation.

How the condition / AID / abend traps work

After every `EXEC CICS` command, the dispatcher writes `EIBRESP` and then consults three optional per-task tables:

Table	Populated by	Consulted	Branches when
Condition map	<code>HANDLE CONDITION</code> / <code>IGNORE CONDITION</code>	After every command	<code>EIBRESP</code> $\neq$ <code>NORMAL</code> and a label is armed for that condition (or for <code>ERROR</code>).
AID map	<code>HANDLE AID</code>	After every command that updates <code>EIBAID</code> (<code>SEND MAP</code> , <code>RECEIVE MAP</code> , <code>RECEIVE</code>)	The new <code>EIBAID</code> matches an armed key (<code>PF1</code> – <code>PF24</code> , <code>PA1</code> – <code>PA3</code> , <code>ENTER</code> , <code>CLEAR</code> , or the catch-all <code>ANYKEY</code>).
Abend exit	<code>HANDLE ABEND</code>	When the task abends	An exit is armed and not cancelled.

When a trap fires, control transfers to the named label as if the program had `GO TO`'d (COBOL) or `SIGNAL`'d (REXX) it directly. The trap *stays armed* across the branch — re-arming after each fire is not required (and the `IGNORE` / bare-name disarm forms exist for when the program wants the trap to stop firing).

SYNCPPOINT

Commit the current unit of work and begin a new one.

Format

EXEC CICS SYNCPPOINT END-EXEC

Description Clears the task's undo journal. All file / TS writes performed since the last SYNC-POINT (or since the task began) are made permanent; they can no longer be rolled back. A fresh, empty journal is started and subsequent mutations accumulate against it.

SYNCPPOINT is implicit on EXEC CICS RETURN — programs that run to completion never need to call it explicitly. The explicit form is useful in long-running tasks that want to checkpoint partial progress, or in programs that mix recoverable phases with phases where rollback is no longer meaningful.

Options None.

Conditions

Condition	EIBRESP	Cause
NORMAL	0	Journal cleared.

SYNCPPOINT cannot fail in bricks because the underlying bbolt writes have already committed by the time the journal entry was appended.

Example

* Phase 1: build the audit row, commit immediately so a later
* validation failure does not undo the audit trail.

```
EXEC CICS WRITEQ TS QUEUE('AUDIT') FROM(WS-LOG) END-EXEC.  
EXEC CICS SYNCPPOINT END-EXEC.
```

* Phase 2: real business work; rolled back if validation fails.

```
EXEC CICS REWRITE FILE('CUSTOMERS') FROM(WS-CUST) END-EXEC.  
IF WS-INVALID  
    EXEC CICS SYNCPPOINT ROLLBACK END-EXEC  
    EXEC CICS ABEND ABCODE('VAL1') END-EXEC  
END-IF.
```

SYNCPPOINT ROLLBACK

Discard the current unit of work and begin a new one.

Format

EXEC CICS SYNCPOINT ROLLBACK END-EXEC

Description Walks the task's undo journal in reverse and applies each inverse operation:

- WRITE is undone by deleting the new key.
- REWRITE is undone by re-writing the captured pre-image.
- DELETE is undone by re-writing the deleted record.
- WRITEQ TS (append) is undone by deleting the appended item.
- WRITEQ TS ... REWRITE is undone by re-writing the pre-image.
- DELETEQ TS is undone by restoring every prior item from a snapshot taken before the delete.

After the walk the journal is cleared and a new unit of work begins.

ROLLBACK is implicit when EXEC CICS ABEND runs **without** a HANDLE ABEND exit. When an exit *does* intercept the abend, the exit code decides whether to call SYNCPOINT ROLLBACK (typical) or SYNCPOINT (commit partial work and continue).

Options ROLLBACK is the only option — it is what distinguishes this form from plain SYNCPOINT.

Conditions

Condition	EIBRESP	Cause
NORMAL	0	All journal entries successfully reverted.
ROLLEDBACK	82	One or more inverse ops failed (the underlying bbolt error is logged). The journal is still cleared; some writes may remain partially undone.

Example

ADDRESS CICS

```
EXEC CICS WRITE FILE('ACCT') RIDFLD(DEBIT_ID) FROM(DEBIT_REC) END-EXEC
EXEC CICS WRITE FILE('ACCT') RIDFLD(CREDIT_ID) FROM(CREDIT_REC) END-EXEC
```

```
IF DEBIT_TOTAL <> CREDIT_TOTAL THEN DO
  EXEC CICS SYNCPOINT ROLLBACK END-EXEC      /* both writes undone */
  EXEC CICS ABEND ABCODE('IMBL') END-EXEC
END
```

```
EXEC CICS SYNCPOINT END-EXEC                /* commit both writes */
```


HANDLE CONDITION

Arm program labels to receive control on named EXEC CICS conditions.

Format

```
EXEC CICS HANDLE CONDITION cond1[(label)] cond2[(label)] ...  
END-EXEC
```

Description Builds a per-task map from condition name to label. After every EXEC CICS command, when EIBRESP $\neq$ NORMAL, the dispatcher consults the map:

1. If the *exact* condition name is armed, control branches to the named label.
2. Otherwise, if the catch-all ERROR is armed, control branches to that label.
3. Otherwise, the response code is left in EIBRESP for the program to test, exactly as if HANDLE CONDITION had never been issued.

The (label) form arms the trap; the bare condition name (no parens) **disarms** it. Arming a previously-armed condition replaces the old label.

HANDLE CONDITION is the legacy CICS error-handling style. Modern programs that use RESP(rc) style (or REXX SIGNAL ON ERROR) do not need it; the two styles can coexist within the same program because both ultimately read EIBRESP.

Options condN(label) Arm condN (one of the names in Chapter 12. Response codes, e.g. NOTFND, DUPREC, MAPFAIL, INVREQ, IOERR, LENGERR, QIDERR, ITEMERR, ENDFILE, PGMIDERR, EOC) so it branches to label.

condN (no parens) Disarm condN.

ERROR(label) Arm a catch-all that fires for any non-NORMAL response not matched by a more specific arm.

A single HANDLE CONDITION may combine any number of arm and disarm options.

Conditions

Condition	EIBRESP	Cause
NORMAL	0	Map updated.
INVREQ	16	Unknown condition name.

Example

```
ADDRESS CICS
```

```
EXEC CICS HANDLE CONDITION NOTFND(NOREC)  
                                DUPREC(DUP)  
                                ERROR(OOPS)
```

```
END-EXEC
```

```
EXEC CICS READ FILE('CUSTOMERS') INTO(REC) RIDFLD(CKEY) END-EXEC
SAY 'Found:' REC
EXIT
```

```
NOREC: SAY 'No customer with key' CKEY; EXIT 4
DUP:   SAY 'Duplicate key on insert'; EXIT 8
OOPS:  SAY 'Other CICS error, RC=' EIBRESP; EXIT 12
```

IGNORE CONDITION

Suppress the HANDLE CONDITION trap for one or more conditions.

Format

```
EXEC CICS IGNORE CONDITION cond1 cond2 ...
END-EXEC
```

Description Marks each named condition as *ignored*: even if HANDLE CONDITION previously armed it, the trap will no longer fire and the program must test EIBRESP itself. IGNORE CONDITION stays in effect until a later HANDLE CONDITION cond(label) re-arms the same condition.

The distinction between *unarmed* (never armed, or disarmed by the bare-name form) and *ignored* matters only when ERROR(label) is armed: an unarmed condition still falls through to ERROR, but an ignored one does not.

Options Each operand is a bare condition name — no parentheses, no label. Multiple conditions may be combined in a single call.

Conditions

Condition	EIBRESP	Cause
NORMAL	0	Map updated.
INVREQ	16	Unknown condition name.

Example

```
EXEC CICS HANDLE CONDITION ERROR(BADIO) END-EXEC.
```

```
* We *expect* NOTFND on this READ — don't trip the ERROR trap.
EXEC CICS IGNORE CONDITION NOTFND END-EXEC.
```

```
EXEC CICS READ FILE('CUSTOMERS') INTO(WS-REC) RIDFLD(WS-KEY) END-EXEC.
IF EIBRESP = 13
  PERFORM CREATE-NEW-CUSTOMER
END-IF.
```

HANDLE AID

Arm program labels to receive control on a particular attention key.

Format

```
EXEC CICS HANDLE AID key1[(label)] key2[(label)] ...  
END-EXEC
```

Description Builds a per-task map from AID byte to label. After any command that updates EIBAID (SEND MAP, RECEIVE MAP, RECEIVE), the dispatcher consults the map and branches if the new EIBAID matches an armed key.

The bare key name (no parens) disarms a previously-armed key. ANYKEY is a catch-all matched by any AID for which no specific arm exists.

Options **keyN(label)** / **keyN** keyN is one of ENTER, CLEAR, PA1–PA3, PF1–PF24, or ANYKEY. Parenthesised arms the trap; bare disarms it.

Conditions

Condition	EIBRESP	Cause
NORMAL	0	Map updated.
INVREQ	16	Unknown AID name.

Example

```
ADDRESS CICS
```

```
EXEC CICS HANDLE AID PF3(QUIT) PF12(QUIT) CLEAR(QUIT) END-EXEC
```

```
DO FOREVER
```

```
    EXEC CICS RECEIVE MAP('CUST1') INTO(SCR.) END-EXEC
```

```
    ... process ...
```

```
    EXEC CICS SEND MAP('CUST1') FROM(SCR.) ERASE END-EXEC
```

```
END
```

```
QUIT:
```

```
    EXEC CICS RETURN END-EXEC
```

HANDLE ABEND

Arm a program-level abend exit.

Format

```
EXEC CICS HANDLE ABEND { LABEL(label) | PROGRAM(name) | CANCEL | RESET }  
END-EXEC
```

Description Registers an exit that runs when the task abends (either via EXEC CICS ABEND or via an unhandled runtime error). The exit *replaces* the implicit SYNCPOINT ROLLBACK that an uncaught abend would perform — it is now the exit's responsibility to call SYNCPOINT or SYNCPOINT ROLLBACK explicitly.

EIBABCODE is set to the abend code (default AAAA) before the exit receives control.

Options **LABEL(label)** Branch to label within the current program when the task abends. This is the form REXX programs almost always use.

PROGRAM(name) XCTL to the named program when the task abends. Currently accepted by the parser; the runtime treats PROGRAM(name) as equivalent to LABEL(name) (i.e. it branches inside the current program rather than transferring control).

CANCEL Disarm the current exit and restore the previous one (one level of nesting is preserved).

RESET Re-enable an exit previously suspended by CANCEL.

Exactly one of LABEL, PROGRAM, CANCEL, RESET must be given.

Conditions

Condition	EIBRESP	Cause
NORMAL	0	Exit registered or restored.
INVREQ	16	Conflicting / missing options, or RESET with no prior exit.

Example

```
ADDRESS CICS
```

```
EXEC CICS HANDLE ABEND LABEL(CLEANUP) END-EXEC
```

```
EXEC CICS WRITE FILE('LEDGER') RIDFLD(K) FROM(REC) END-EXEC
```

```
EXEC CICS WRITE FILE('AUDIT') RIDFLD(K) FROM(REC) END-EXEC
```

```
/* something later may ABEND */  
EXIT
```

```
CLEANUP:
```

```
    SAY 'Caught abend' EIBABCODE
```

```
    EXEC CICS SYNCPOINT ROLLBACK END-EXEC          /* undo both writes */
```

```
    EXEC CICS RETURN END-EXEC
```

Chapter 11. The Execute Interface Block (EIB)

The EIB is a per-task scratch area populated by the dispatcher before the program runs and by each EXEC CICS command on return. In bricks the EIB is exposed as a set of well-known variable names in the program's frame; both REXX and COBOL auto-inject the names that aren't already declared.

Field	Set by	Meaning
EIBAID	SEND MAP / RECEIVE MAP	Single-byte AID character of the most recent map response. Use C2X(EIBAID) (REXX) or compare with X'F3' (COBOL) to detect PF/PA/CLEAR/ENTER.
EIBCP0SN	SEND MAP / RECEIVE MAP	1-based cursor position of the most recent map response.
EIBCALEN	dispatcher (per-task entry)	Length of DFHCOMMAREA flowed in from the prior task or LINK caller. Zero when no COMMAREA was passed.
EIBTRMID	dispatcher	This terminal's id. Same as ASSIGN TERMD(...).
EIBRESP	every EXEC CICS	The response code of the most recent command (see Chapter 12).
EIBRESP2	every EXEC CICS	The secondary response code of the most recent command (currently always 0).
RC	every EXEC CICS	Mirror of EIBRESP, for REXX IF RC <> 0 style.
DFHCOMMAREA	dispatcher / LINK / RETURN	The COMMAREA bytes; in COBOL, auto-injected as PIC X(2000) if not declared.

REXX

EIBAID and friends are ordinary REXX variables that the handlers write through the cics.Frame interface. A REXX program tests them exactly like any other variable:

```
IF EIBRESP <> 0 THEN SAY 'Command failed, RC=' EIBRESP
IF C2X(EIBAID) = 'F3' THEN EXEC CICS RETURN END-EXEC
```

SIGNAL ON ERROR (Chapter 18) is the alternative to per-call IF EIBRESP <>.

COBOL

The same names are auto-injected as PIC items if the program doesn't declare them (`cobol.ensureSystemItems`). Test them with ordinary IF clauses:

```
IF EIBRESP NOT = 0
    DISPLAY 'Command failed'
END-IF.

IF EIBAID = X'F3'
    EXEC CICS RETURN END-EXEC
END-IF.
```

There is no COBOL equivalent of `SIGNAL ON ERROR` yet.

Chapter 12. Response codes

After every `EXEC CICS` command the handler writes `EIBRESP`, `EIBRESP2`, and `RC` into the program's frame. The full set of constants is in `cics/resp.go`; the values a typical bricks program tests for are:

Constant	Value	Meaning
NORMAL	0	Success.
ERROR	1	Generic error.
EOC	6	End-of-chain (second RECEIVE in same task).
NOTFND	13	Record / key not found.
DUPREC	14	WRITE collided with an existing key.
DUPKEY	15	Duplicate key on a non-unique index (reserved).
INVREQ	16	Invalid request: bad option combination, missing data store, invalid name.
IOERR	17	Underlying store / network IO failed.
NOSPACE	18	Out of storage (reserved).
NOTOPEN	19	File not open (reserved).
ENDFILE	20	Browse walked past the last key (or out of GENERIC prefix).
LENGERR	22	Length mismatch.
QZERO	23	Queue is empty (reserved).
ITEMERR	26	TS item out of range; typical at end-of-queue.

Constant	Value	Meaning
PGMIDERR	27	LINK / XCTL target not in transactions.conf, or sub-program errored, or caller is not authorised for the target.
MAPFAIL	36	SEND MAP / RECEIVE MAP could not find or render the named map.
QIDERR	44	TS queue id invalid or unknown.

Idiomatic error handling

REXX, per-call test:

```
EXEC CICS READ FILE('CUSTOMERS') INTO(REC) RIDFLD(K) END-EXEC
SELECT
  WHEN EIBRESP = 0 THEN NOP
  WHEN EIBRESP = 13 THEN MSG = 'Customer' K 'not found'
  OTHERWISE          MSG = 'I/O error, RC=' EIBRESP
END
```

REXX, signal-on:

```
SIGNAL ON ERROR NAME CICSERR
EXEC CICS READ FILE('CUSTOMERS') INTO(REC) RIDFLD(K) END-EXEC
...
EXIT
CICSERR:
  SAY 'CICS error at line' SIGL ', RC=' RC
  EXIT
```

COBOL:

```
EXEC CICS READ FILE('CUSTOMERS') INTO(REC) RIDFLD(CKEY) END-EXEC
EVALUATE EIBRESP
  WHEN 0    CONTINUE
  WHEN 13  MOVE 'Customer not found' TO MSG
  WHEN OTHER MOVE 'I/O error' TO MSG
END-EVALUATE.
```

Chapter 13. Commands not implemented

The following CICS commands are recognized by the parser but explicitly rejected; programs that use them will receive INVREQ with a clear error message rather than silent misbehaviour.

Command	Why
READQ TD, WRITEQ TD, DELETEQ TD	Transient data has one-shot read-and-destroy semantics that TS doesn't; silently routing TD to TS would mask a real difference.

The following commands are not implemented and the parser does not recognize them at all:

Command	Use
START, RETRIEVE GETMAIN, FREEMAIN	Asynchronous transaction starting. Dynamic storage. REXX has dynamic variables; COBOL has WORKING-STORAGE.
ENQ, DEQ	User-level resource locking. File-level locking already exists via READ ... UPDATE.

Part 3. The REXX language

Chapter 14. REXX program structure

A bricks REXX program is a flat sequence of statements with optional labels and procedures. Execution begins at the first statement; the program ends when execution falls off the bottom, or hits EXIT.

```
/* HEL0 - minimum REXX program */
ADDRESS CICS
EXEC CICS SEND MAP('HEL01') ERASE END-EXEC
EXEC CICS RETURN END-EXEC
```

Procedures

A label followed by PROCEDURE [EXPOSE list] defines a procedure with its own variable scope. EXPOSE re-routes named variables to the caller's frame recursively.

```
CALL GREET 'Alice'
EXIT
```

```
GREET: PROCEDURE EXPOSE LANG.
  PARSE ARG NAME
  SAY LANG.HELLO NAME
  RETURN
```

ADDRESS

ADDRESS <env> switches the active command handler. Bare strings inside an ADDRESS scope are commands routed to that handler; bricks ships a CICS handler.


```
ADDRESS CICS
"SEND MAP('CUST1') FROM(SCR.) ERASE"
"RETURN"
```

Chapter 15. Variables and stems

- **Simple variables** are case-insensitive. An unset variable resolves to its uppercased name (REXX NOVALUE convention).

- **Stems** have a default value plus per-tail values:

```
STEM. = 'unset'
STEM.42 = 'forty-two'
SAY STEM.1 /* unset */
SAY STEM.42 /* forty-two */
```

- **Compound-variable tail substitution.** Non-numeric tail symbols are resolved at every reference. With J = 3, the symbol A.J reads or writes A.3. Pure numeric tails (A.0, A.42) and unset tail symbols (REXX NOVALUE) remain literal. Multi-segment tails work too: with I=1, J=2, A.I.J references A.1.2.
- **DROP name [name...]** removes one or more variables. A trailing . drops the entire stem (default + every tail) — useful for resetting an accumulator between paginated reads: DROP RECS..

See Appendix B for the canonical compound-symbol pitfall.

Chapter 16. Control flow

Construct	Forms
IF expr THEN [ELSE] SELECT ... WHEN ... OTHERWISE ... END DO, DO N, DO var=a TO b BY s, DO WHILE, DO UNTIL, DO FOREVER DO var OVER stem.	one-line or block OTHERWISE NOP works for the empty branch the usual loop family
LEAVE [ctrlvar] ITERATE [ctrlvar]	iterate over each tail of a stem; numeric tails sort first, then lexicographic exit the innermost (or named outer) DO skip to the next iteration of the innermost (or named outer) DO
CALL, RETURN, EXIT SIGNAL <label> INTERPRET expr	procedure call, return, exit non-local jump evaluate the string value of expr as REXX source and execute
NUMERIC DIGITS n / FUZZ n / FORM SCIENTIFIC ENGINEERING NOP	basic settings honoured; arithmetic is float64 internally a real no-op statement

Chapter 17. PARSE templates

PARSE [UPPER] {VAR var | VALUE ... WITH | ARG | PULL} template

Template features supported:

- String anchors ('literal').
- Absolute column markers (n).
- Relative column markers (+n / -n).
- The . placeholder (skip a token).
- Bare variable runs.

```
PARSE VAR LINE  TID  CKEY  .                /* whitespace tokens */
PARSE VAR REC   NM  '|' AD  '|' CY  '|' PH    /* '|' delimiter      */
PARSE VAR ROW    1 NAME 21 ADDR 51 PHONE     /* fixed columns      */
```

Chapter 18. Conditions and SIGNAL ON

SIGNAL ON {ERROR | NOVALUE | SYNTAX | HALT} [NAME label] arms a condition. When it fires:

- SIGL is set to the source line of the failing statement.
- Control jumps to the labelled handler.

SIGNAL OFF cond disarms.

Condition	Fires on
ERROR	An EXEC CICS command returns a non-zero RC.
NOVALUE	A reference to an unset simple or compound variable.
SYNTAX	Any other interpreter error — bad numeric, divide by zero, unknown function, etc.
HALT	An external halt request.

Example

```
SIGNAL ON ERROR  NAME CICSERR
SIGNAL ON SYNTAX NAME OOPS
```

```
ADDRESS CICS
EXEC CICS READ FILE('CUSTOMERS') INTO(REC) RIDFLD(CKEY) END-EXEC
EXIT
```

```
CICSERR:
  SAY 'EXEC CICS failed at line' SIGL ', RC=' RC
```

EXIT 12

OOPS:

SAY 'Interpreter error at line' SIGL

EXIT 16

Without an armed trap, the legacy “test EIBRESP after every verb” pattern still works. For per-condition (rather than blanket) traps, use EXEC CICS HANDLE CONDITION (Chapter 10); the two styles can coexist.

Chapter 19. Built-in functions

Family	Functions
Length / index	LENGTH, POS, LASTPOS, WORDS, WORDPOS, WORDINDEX, WORDLENGTH, COUNTSTR
Substring	SUBSTR, LEFT, RIGHT, SUBWORD, WORD, DELSTR, DELWORD, INSERT, OVERLAY, CHANGESTR
Whitespace / case	STRIP, SPACE, CENTER (alias CENTRE), JUSTIFY, UPPER, LOWER, REVERSE, COPIES, ABBREV
Translation	TRANSLATE, VERIFY, COMPARE, X RANGE, BITAND, BITOR, BITXOR
Conversion	C2X, X2C, D2X, X2D, D2C, C2D, B2X, X2B
Numeric	ABS, MAX, MIN, INT, TRUNC, MOD (= // operator), SIGN, FORMAT(num, before, after), DIGITS, FUZZ, FORM
Type / data	DATATYPE (with N, W, A options), LENGTH
Date / time	DATE (N/S/E/U/O/B; B does basedate arithmetic), TIME
Variables / args	VALUE, ARG, RANDOM, ERRORTXT

VALUE accepts the canonical 1-arg read form (VALUE('SCR.ROW' || J)) and the 2-arg assignment form (CALL VALUE 'SCR.ROW' || J, LINE — sets the variable named at runtime and returns the prior value).

C2X is the standard way to compare EIBAID to a PF-key code: IF C2X(EIBAID) = 'F7' THEN ... for PF7.

DATE('B') and DATE('B', 'YYYYMMDD', 'S') give days since 0001-01-01 — subtract two basedates for an exact day delta.

Operators

+ − ∗ / % // ∗∗, comparisons (numeric when both sides parse as numbers, trimmed-string otherwise), || and juxtaposition concat, & |, unary \.

Part 4. The COBOL language

Chapter 20. COBOL source format

Bricks ships a free-form COBOL interpreter (`cobol/`) that sits beside REXX as a second front end on the same EXEC CICS surface. The language-specific layer is `cobol/frame.go`, which adapts COBOL's group-item world to the REXX-style STEM.TAIL lookup the CICS handlers use.

Source rules

- **Free-form.** No column 1-72 ruling; no Area A / Area B.
- **Hyphens** are allowed in identifiers (CUST-RECORD).
- **Comments** use modern `*>` anywhere on a line, or legacy `*` in column 1.
- **Strings** use `'...'` or `"..."` with quote-doubling for embedded quotes.
- **Hex literals** like `X'F3'` and `X"7C"` decode to their byte value — used for `IF EIBAID = X'F3'` to check PF3 / PF12 / etc. without `C2X(EIBAID) = 'F3'` round-trips.

Divisions

The four divisions parse in their canonical order:

```
IDENTIFICATION DIVISION.  
PROGRAM-ID. HELC.
```

```
ENVIRONMENT DIVISION.          *> optional, must be empty
```

```
DATA DIVISION.  
WORKING-STORAGE SECTION.  
01 SCR.  
   05 INFOLINE PIC X(78).  
   05 GREETING PIC X(20).
```

```
PROCEDURE DIVISION.  
MAIN.  
   MOVE 'Hello' TO GREETING.  
   EXEC CICS SEND MAP('HEL01') FROM(SCR) ERASE END-EXEC.  
   EXEC CICS RETURN END-EXEC.  
   STOP RUN.
```

LINKAGE SECTION is not yet parsed; DFHCOMMAREA is auto-injected as `PIC X(2000)` if the program doesn't declare it (see `cobol.ensureSystemItems`), so a sub-program can `MOVE DFHCOMMAREA TO key` immediately. The dispatcher strips trailing space when reading the COBOL frame's DFHCOMMAREA back out, so a fixed-width buffer round-trips cleanly across an inter-language EXEC CICS LINK.

Chapter 21. DATA DIVISION

- **PIC clauses:** X(n) alphanumeric, 9(n) integer, S9(n) signed, 9(n)V99 decimal (the V is positional, no real binary scaling yet — arithmetic is float64 internally).
- **VALUE:** VALUE 'literal', VALUE 42, VALUE SPACES, VALUE ZEROS, VALUE HIGH-VALUES, VALUE LOW-VALUES, VALUE QUOTES.

- **Group items:**

```
01 PARENT.  
    05 CHILD PIC X(8).  
    05 OTHER PIC X(4).
```

Children are stored as offsets into a single parent buffer, so MOVE to the parent fans out and EXEC CICS SEND MAP FROM(PARENT) walks the children for field values.

- **Data names can be reused across groups.** A child name that appears under more than one group is fine; the parser accepts the declaration and tracks the collision in Program.AmbiguousNames. Unqualified access to such a name is a runtime error (“ambiguous reference”) — disambiguate with OF (or its synonym IN):

```
01 SCR.  
    05 CUSTNO PIC X(8).  
01 DET.  
    05 CUSTNO PIC X(8).  
...  
MOVE 'A1234567' TO CUSTNO OF SCR.  
MOVE 'B7654321' TO CUSTNO OF DET.  
DISPLAY CUSTNO OF SCR.
```

Single-step qualification (X OF Y) picks the matching child anywhere in Y’s subtree; multi-step (X OF Y OF Z) chains through nested groups. Bare names that are unique across the program continue to work without qualification — the pre-7 runtime/cobol/gust.cob convention of prefixed child names (DCUSTNO, DNAME, DMSG) still compiles and runs, it just isn’t required any more.

A sibling-duplicate (two children of the SAME parent sharing a name) is still a parse-time error — no qualifier can disambiguate between siblings of one group.

Chapter 22. PROCEDURE DIVISION

Statements supported

MOVE, DISPLAY, STOP RUN, GOBACK, EXIT, EXIT PROGRAM, CONTINUE (no-op), IF ... [ELSE] ... END-IF, EVALUATE subject WHEN value [WHEN value] ... [WHEN OTHER] ... END-EVALUATE, PERFORM para, PERFORM para UNTIL cond, PERFORM para N TIMES, PERFORM para VARYING idx FROM x BY y UNTIL cond, GO TO para, COMPUTE target [ROUNDED] = expr [ON SIZE ERROR ... END-COMPUTE], ADD a TO b [GIVING c] [ROUNDED] [ON SIZE ERROR ... END-ADD], SUBTRACT a FROM b [GIVING c] [ROUNDED] [ON SIZE ERROR ...

END-SUBTRACT], MULTIPLY a BY b [GIVING c] [ROUNDED] [ON SIZE ERROR ... END-MULTIPLY], DIVIDE a INTO b [GIVING c] [ROUNDED] [ON SIZE ERROR ... END-DIVIDE], DIVIDE a BY b GIVING c [ROUNDED] [ON SIZE ERROR ... END-DIVIDE], STRING ... DELIMITED BY (SIZE | 'lit') INTO target END-STRING, UNSTRING source DELIMITED BY 'lit' INTO t1 t2 ... END-UNSTRING, INSPECT subject TALLYING counter FOR (ALL | LEADING | CHARACTERS) [needle] [BEFORE/AFTER INITIAL delim], INSPECT subject REPLACING (ALL | LEADING | FIRST | CHARACTERS) [needle] BY replacement [BEFORE/AFTER INITIAL delim], EXEC CICS ... END-EXEC.

Every target name above (the second operand of MOVE, the target of COMPUTE / ADD / SUBTRACT / MULTIPLY / DIVIDE / GIVING / STRING INTO / UNSTRING INTO / INSPECT, and the subject of EVALUATE / IF) accepts qualification via OF / IN (see the Data Division section on globally non-unique names) and subscripts via (idx) for OCCURS-resident items.

ROUNDED and ON SIZE ERROR

ROUNDED applied to COMPUTE and the four arithmetic verbs rounds the final value half-away-from-zero to the destination's PIC scale instead of truncating toward zero (the unROUNDED default).

ON SIZE ERROR runs its statement list when the calculation overflows the int64 fixed-point engine, the result's integer-part digits don't fit in the target PIC, or a DIVIDE divisor is zero; when present, the destination is left untouched on a size-error firing. Without an ON SIZE ERROR branch the value is silently truncated/wrapped as before so existing programs aren't disrupted.

PERFORM N TIMES and PERFORM VARYING

PERFORM FILL-ROW 15 TIMES.

```
PERFORM FILL-ROW
    VARYING I FROM 1 BY 1 UNTIL I > 15.
```

PERFORM ... N TIMES runs the named paragraph exactly N times; N can be a numeric literal or a numeric data item. PERFORM ... VARYING idx FROM x BY y UNTIL cond initialises idx to x, runs the body while cond is false, then increments idx by y between iterations.

OCCURS arrays

```
01 TABLE-AREA.
    05 ROW OCCURS 15 TIMES.
        10 K PIC X(8).
        10 N PIC X(28).
01 I PIC 9(4).
...
MOVE 'KEY-A' TO K(1).
MOVE 'KEY-B' TO K(I).
PERFORM SHOW-ROW VARYING I FROM 1 BY 1 UNTIL I > 15.
```

OCCURS n TIMES declares n copies of an item; references add a parenthesised 1-based subscript expression that picks one. The subscript can be any numeric expression — a literal, a data item, or arithmetic — and is bounds-checked at runtime. Subscripts work on group items (ROW(5)) and

on elementaries inside the OCCURS group (K(5)) alike. The cap is 4096 occurrences per item; nested OCCURS (an OCCURS group inside another OCCURS group) is rejected at parse time.

Periods

Periods are scope-terminators, not per-statement: a statement inside an IF ... END-IF body has no period; only the unscoped statement at the paragraph level does.

Postfix NOT

IBM-style postfix NOT ($X \text{ NOT} = Y$, $X \text{ NOT} > Y$) is supported in IF / EVALUATE / PERFORM UNTIL conditions.

Intrinsic functions

The shipped intrinsic library is:

Function	Args	Result
FUNCTION UPPER-CASE(s)	1 alphanumeric	s uppercased.
FUNCTION LOWER-CASE(s)	1 alphanumeric	s lowercased.
FUNCTION TRIM(s)	1 alphanumeric	s with leading and trailing spaces removed (both ends; one-arg form).
FUNCTION REVERSE(s)	1 alphanumeric	s reversed (rune-aware).
FUNCTION LENGTH(s)	1 alphanumeric	numeric byte length of the operand's storage (PIC X(8) □ 8).
FUNCTION NUMVAL(s)	1 alphanumeric	numeric value of s after stripping leading / trailing whitespace; errors on non-numeric content (does not silently coerce to zero).
FUNCTION POS(needle, haystack)	2 alphanumeric	1-based byte position of needle in haystack, or 0 when not found. Mirrors REXX's POS(). Empty needle returns 0.

Numeric-returning intrinsics (LENGTH, NUMVAL, POS) participate in IF / EVALUATE / arithmetic comparisons as numeric operands, so IF FUNCTION POS(NEEDE, HAY) > 0 works without an intermediate COMPUTE. The remaining intrinsics return alphanumeric and compare by bytes (rstripped).

FUNCTION POS is the substring-search hook used by runtime/cobol/gusl.cob to filter the customers file when the operator types a search term at GUST's S action.

INSPECT

```
INSPECT subject TALLYING counter FOR (ALL | LEADING | CHARACTERS)
                                   [needle] [BEFORE/AFTER INITIAL delim]
                                   [, ...]
INSPECT subject REPLACING (ALL | LEADING | FIRST | CHARACTERS)
                           [needle] BY replacement
                           [BEFORE/AFTER INITIAL delim]
                           [, ...]
```

Quantifiers:

- **ALL** — every occurrence of `needle` (TALLYING) or every occurrence rewritten (REPLACING).
- **LEADING** — only successive occurrences at the start of the in-scope region.
- **FIRST** — only the first occurrence (REPLACING only).
- **CHARACTERS** — every character in the in-scope region; for TALLYING this just counts characters, for REPLACING it overwrites each one with the first byte of `replacement`.

Limiters:

- **BEFORE INITIAL `delim`** narrows the in-scope region to everything before the first occurrence of `delim`.
- **AFTER INITIAL `delim`** narrows to everything after the first occurrence of `delim`.

Both can be set on the same phrase; AFTER picks the start, BEFORE picks the end.

Multiple phrases can be combined per statement, optionally separated by commas:

```
INSPECT REC REPLACING ALL ',' BY '|' ALL ';' BY '/'.
INSPECT REC TALLYING C FOR LEADING '0', ALL 'X' AFTER INITIAL ' '.
```

Counter behaviour: TALLYING **accumulates** — successive INSPECT TALLYING statements add to the existing counter value rather than resetting, matching IBM semantics. REPLACING phrases run left-to-right in declared order, so later phrases see the output of earlier ones.

Subject, counter, and operands all accept qualified data-names (INSPECT REC OF DET ...). Trailing PIC X padding is stripped from both the subject and the needle before matching, so a 30-byte filter containing F00 plus trailing spaces matches against the meaningful content of the subject rather than against the padding.

IBM's CONVERTING form is not parsed yet.

EVALUATE limitations

EVALUATE only supports the simple value form. EVALUATE TRUE / EVALUATE FALSE is rejected at parse time with a hint to rewrite as IF/ELSE — those forms (and WHEN ... THRU ... ranges, multi-subject ALSO clauses, condition-name arms) are deferred.

Chapter 23. The EIB block in COBOL

EIBRESP, EIBRESP2, EIBAID, EIBCP0SN, EIBCALEN, EIBTRMID, RC, and DFHCOMMAREA are auto-injected if not declared, so most programs need no boilerplate. After every EXEC CICS, EIBRESP and EIBRESP2 are populated by the handler the same way they are for REXX. The legacy “test EIBRESP after every verb” pattern is the most common idiom; for non-local error handling without a REXX-style SIGNAL ON ERROR, use EXEC CICS HANDLE CONDITION / EXEC CICS HANDLE ABEND (Chapter 10).

See Chapter 11 for the field meanings.

Chapter 24. EXEC CICS in COBOL

The COBOL parser collects every token between EXEC CICS and END-EXEC and reconstructs the body verbatim for the same `cics.ParseCommand` REXX uses. Three consequences:

- **Map-field names must match group-child names exactly.** When EXEC CICS SEND MAP('CUST1') FROM(SCR) fires, the CICS handler asks the frame for SCR.INFOLINE, SCR.MSG, etc. — so SCR must have children spelled exactly as the map declares fields. Real COBOL solves this with BMS-generated copybooks; bricks doesn't ship those, so the operator declares fields by hand.
- **DFHCOMMAREA is the COMMAREA marshalling slot.** Caller passes bytes in via the frame; sub-program reads with MOVE DFHCOMMAREA TO ..., mutates a working copy, and MOVE ... TO DFHCOMMAREA. Trailing space is stripped by the dispatcher on the way back out.
- **Command-line arguments arrive via EXEC CICS RECEIVE INTO(...).** When the operator types EXAM 1 2 3 at the blank prompt, the dispatcher hands the unedited line (TRANSID prefix included) to the first task in the chain. The program reads it with EXEC CICS RECEIVE INTO(WS-INPUT) LENGTH(WS-LEN) END-EXEC and parses with UNSTRING WS-INPUT DELIMITED BY ' ' INTO TRANSID A B C END-UNSTRING. Single-shot per task, then EOC. See runtime/cobol/exam.cob.

EXEC CICS SEND TEXT FROM(area) [ERASE] works the same way. Reconstruction routes through the shared `cics.Handler`, so any verb REXX can issue, COBOL can issue too. The body is laid out as a flat row-major buffer (every cols bytes = one row, default 80) — 3270 has no LF, so build area as a group of PIC X(80) children and fill them row by row, or compose a single PIC X(n*80) with a matching MOVE per row.

The full EXEC CICS verb set is documented in Part 2; behaviour is identical between REXX and COBOL.

Chapter 25. Restrictions and deferred features

Disallowed

- **Calculated GOTOs.** GO TO DEPENDING ON is rejected at parse time. The bricks runtime gives every task its own heap and stack with no static control-block aliasing; calculated GO-

TOs would require a per-program label-table the parser deliberately doesn't build.

Deferred Syntax Coverage

- Reference modification (DATE-FIELD(1:4) for substring access).
- Multi-dimensional OCCURS. One-level OCCURS is supported (see Chapter 22); the parser rejects an OCCURS item whose chain already contains another OCCURS ancestor.
- SCREENHT-based map family suffix (e.g. CUST1L on a mod-4 screen). REXX programs do this with a runtime IF SCRH >= 43 THEN ... fallback after a MAPFAIL; the COBOL twins always render the unsuffixed mod-2 maps for now.

Part 5. Sample programs

Chapter 26. Pre-installed sample transactions

COBOL

Transid	File	Notes
HELC	runtime/cobol/hello.cob	Hello-world; SEND MAP('HEL01') FROM(SCR), RETURN. Smallest end-to-end demo.
QAGC	runtime/cobol/qagc.cob	COBOL twin of QAGR (REXX). Validates the QAGE1 birthdate, computes age in years and approximate days, sends QAGR1. Pseudo-conversational redisplay of QAGE1 on validation errors.
GUST	runtime/cobol/gust.cob	COBOL CUST. A=Add, Q=Query, U=Update, D=Delete, L=List, S=Search; the S action LINKs to GUSL with the search term in COMMAREA and renders the match count returned.
GUSV	runtime/cobol/gusv.cob	COBOL twin of CUSV. Validates the customer-number COMMAREA (LINK target).

Transid	File	Notes
GUSL	runtime/cobol/gusl.cob	COBOL twin of CUSL. Renders the customers file via STARTBR / READNEXT / ENDBR. Blank inbound COMMAREA = paginated all-records list (PF7/PF8). Non-blank inbound COMMAREA = filtered single-page browse: every record is FUNCTION POS-tested against the upper-cased filter, the first 15 matches populate ROW1..ROW15, and the total match count flows back through DFHCOMMAREA so the caller (GUST) can render a summary.
EXAM	runtime/cobol/exam.cob	Worked example of reading the operator's command-line arguments. Type EXAM 1 2 3 at the blank prompt.

REXX (selected)

Transid	File	Notes
HEL0	runtime/rexx/hello.rexx	Hello-world with system-info pane on mod 4.
CUST	runtime/rexx/cust.rexx	Full customer maintenance — Add / Query / Update / Delete / List / Search; LINKs to CUSV for validation.
CUSV	runtime/rexx/cusv.rexx	Validation sub-program; called by CUST via EXEC CICS LINK.
CUSL	runtime/rexx/cusl.rexx	Paginated customer list using STARTBR / READNEXT / ENDBR; adapts paging to mod 2 vs mod 4.
QAGE / QAGR	runtime/rexx/qage.rexx / qagr.rexx	Pseudo-conversational chain; QAGE prompts for a birthdate and chains to QAGR to render the result.

Transid	File	Notes
PROD / CONS	runtime/rexx/prod.rexx / cons.rexx	TS queue producer / consumer pair. Conversational; PF3 to exit.
GETC	runtime/rexx/getc.rexx	RECEIVE of command-line + READ FILE + SEND TEXT (no map).

Run any TRANSID by typing it at the blank prompt after CSSN sign-on. Refer to run-time/transactions.conf for the full list and ACL configuration.

Chapter 27. Worked examples

A. Producer / consumer over a TS queue

PROD writes one item per ENTER from an interactive map; CONS reads with the implicit per-task cursor. PF4 in CONS deletes the queue; PF5 rewinds the cursor by chaining back to itself (RETURN TRANSID('CONS')), so the dispatcher's task-end hook clears the cursor.

```
/* PROD: write one item per ENTER */
ADDRESS CICS
DO FOREVER
  EXEC CICS SEND      MAP('PROD1') FROM(SCR.) ERASE END-EXEC
  EXEC CICS RECEIVE MAP('PROD1')                END-EXEC
  IF C2X(EIBAID) = 'F3' THEN EXEC CICS RETURN END-EXEC
  EXEC CICS WRITEQ TS QUEUE(MAP.QNAME) FROM(MAP.PAYLOAD) END-EXEC
END

/* CONS: cursor-advancing read */
ADDRESS CICS
DO FOREVER
  EXEC CICS SEND      MAP('CONS1') FROM(SCR.) ERASE END-EXEC
  EXEC CICS RECEIVE MAP('CONS1')                END-EXEC
  IF C2X(EIBAID) = 'F3' THEN EXEC CICS RETURN END-EXEC
  EXEC CICS READQ TS QUEUE(QNM) INTO(REC) ITEM(GOTI) END-EXEC
  IF EIBRESP = 26 THEN /* ITEMERR - end of queue */ NOP
END
```

B. SEND TEXT + RECEIVE — no BMS map

GETC takes a customer number from the operator's command line (GETC 100), reads the customers KSDS, and paints the record as free-form text. Because **3270 has no LF**, the body is built as a flat row-major buffer — each logical line padded to the 80-column screen width with LEFT(s,80).

```
/* GETC: lookup-and-display, no map */
ADDRESS CICS
```

```

EXEC CICS RECEIVE INTO(BUF) END-EXEC          /* "GETC 100" */
PARSE VAR BUF TID CKEY .
EXEC CICS READ FILE('customers') INTO(REC) RIDFLD(CKEY) END-EXEC
PARSE VAR REC NM '|' AD '|' CY '|' PH
TXT = LEFT('Customer #' || CKEY, 80)          /* row 0 */
TXT = TXT || LEFT(' ', 80)                    /* row 1 (blank) */
TXT = TXT || LEFT('Name:   ' || NM, 80)       /* row 2 */
TXT = TXT || LEFT('Address: ' || AD, 80)      /* row 3 */
EXEC CICS SEND TEXT FROM(TXT) ERASE END-EXEC
EXEC CICS RETURN END-EXEC

```

The COBOL companion runtime/cobol/exam.cob shows the same RECEIVE INTO pattern with UNSTRING ... DELIMITED BY ' ' doing the tokenisation:

```

EXEC CICS RECEIVE INTO(WS-INPUT) LENGTH(WS-LEN) END-EXEC
UNSTRING WS-INPUT DELIMITED BY ' '
    INTO WS-TID WS-A WS-B WS-C
END-UNSTRING.

```

C. Browse with GENERIC prefix

A paginated customer list filtered by a 3-character key prefix:

```

EXEC CICS STARTBR FILE('CUSTOMERS')
                RIDFLD('NY-')
                GENERIC KEYLENGTH(3)
END-EXEC
DO FOREVER
    EXEC CICS READNEXT FILE('CUSTOMERS') INTO(REC) RIDFLD(K) END-EXEC
    IF EIBRESP = 20 THEN LEAVE          /* ENDFILE */
    CALL ADD_ROW K, REC
END
EXEC CICS ENDBR FILE('CUSTOMERS') END-EXEC

```

D. Synchronous LINK with COMMAREA

CUST LINKs to CUSV to validate a customer number:

```

SAV = CKEY                                     /* in: number to validate */
EXEC CICS LINK PROGRAM('CUSV') COMMAREA(SAV) END-EXEC
PARSE VAR SAV STATUS '|' MSG                  /* out: status, message */
IF STATUS <> 'OK' THEN ...

```

CUSV reads DFHCOMMAREA, runs its check, and writes the result back into DFHCOMMAREA before RETURN. The dispatcher hands the final value back to the caller's SAV.

Appendix A. Adapting to terminal size (mod 2 vs mod 4)

A 3270 connection negotiates one of several screen models — typically mod 2 (24 - 80) or mod 4 (43 - 80). Bricks captures the size from the telnet/3270 handshake into `session.TCB.Rows/Cols`, and exposes it to programs via `EXEC CICS ASSIGN`:

```
EXEC CICS ASSIGN SCREENHT(SCRH) SCREENWD(SCRW)
                ALTSCRNHT(AH) ALTSCRNWD(AW) END-EXEC
```

`SEND MAP` always passes the negotiated `DevInfo` through to `go3270.ScreenOpts.AltScreen`, so the underlying datastream uses Erase/Write Alternate (0x7e) and the terminal clears its full buffer — but a 24-row map painted on a 43-row screen leaves rows 24-42 blank. To use the extra real estate the program has to dispatch to a sized map variant.

Convention

Author one map per model. The mod-2 map keeps its bare name; bigger models add a single-letter suffix:

Model	Suffix	Example map names
mod 2	(none)	HEL01, CUST1, CUST2, CUSTL
mod 3	M	HEL01M, CUST1M, ...
mod 4	L	HEL01L, CUST1L, CUST2L, CUSTLL
mod 5	W	HEL01W, ...

The REXX program builds the suffixed name once and dispatches:

```
EXEC CICS ASSIGN SCREENHT(SCRH) END-EXEC
SUFFIX = ''
IF SCRH >= 43 THEN SUFFIX = 'L'
ELSE IF SCRH >= 32 THEN SUFFIX = 'M'

EXEC CICS SEND MAP('HEL01' || SUFFIX) FROM(SCR.) ERASE END-EXEC
IF EIBRESP = 36 THEN DO /* MAPFAIL – sized variant missing */
    EXEC CICS SEND MAP('HEL01') FROM(SCR.) ERASE END-EXEC
END
```

Three properties make this work without any DSL changes:

1. **Same field names across the family.** `hel01.map` and `hel01l.map` both declare `INFO–LINE`, `GREETING`, `FOOTER`. The same `SCR.` stem feeds either one. Bonus tails (e.g. `INF01 / INF02 / ACT1 ...`) are silently ignored on the smaller map (the renderer only writes values for fields that the map declares).
2. **MAPFAIL fallback.** If the suffixed map isn't on disk, the `SEND` returns `EIBRESP = 36` and the program retries with the bare name — so an operator who deletes `hel01l.map` doesn't break mod-4 connections; they just see the 24-80 layout.
3. **Paging arithmetic adapts at runtime.** `custl.rexx` reads `SCREENHT` and uses `ROWS_PER_PAGE = 35` on mod 4 vs 15 on mod 2, then picks `CUSTLL` vs `CUSTL` accordingly.

Bricks ships sized variants for every map the demo transactions use:

Mod-2 map (24-80)	Mod-4 sibling (43-80)
runtime/map/helo1.map	runtime/map/helo1l.map — adds system-information + recent-activity panes
runtime/map/cust1.map (menu)	runtime/map/cust1l.map — adds recent-activity history
runtime/map/cust2.map (detail)	runtime/map/cust2l.map — adds an audit-log pane
runtime/map/custl.map (15-row list)	runtime/map/custll.map (35-row list)

To see the difference: connect with `c3270 -model 2 localhost 2300` vs. `c3270 -model 4 localhost 2300`, sign on, and run HEL0 or CUST. The mod-4 view fills the bottom three-quarters of the screen with extra panels.

Appendix B. Pitfalls and idioms

REXX compound-symbol pitfall

STEM.tail with tail an *unset* symbol resolves to STEM.<TAIL> (a literal tail). With tail a *set* symbol it resolves to STEM.<value-of-tail> — so reusing a map field name (OUT.BIRTH = ... when BIRTH is also a local variable) silently writes the wrong tail.

The convention used by runtime/rexx/cust.rexx is to give locals distinct names from map fields (AKT vs ACTION, CKEY vs CUSTNO, BSTR / NDAYS vs BIRTH / DAYS).

COBOL data names across groups

A child name reused across groups is allowed; bare references that match more than one declaration must be qualified with OF / IN (MOVE x OF DET TO y OF SCR). Bare names that are unique across the program continue to work without qualification. Sibling duplicates within a single group are still rejected at parse time because no qualifier can disambiguate them.

3270 has no LF

SEND TEXT lays its body out as a flat row-major buffer, every cols bytes wrapping to the next row. Programs that expect newline semantics must pad each logical line to the column width (LEFT(s,80) in REXX, PIC X(80) group children in COBOL) and concatenate.

Reset stems before paginated reads

A REXX STEM. accumulator that's reused across pages will leak values from the previous page if not dropped. Use DROP RECS. (with the trailing dot) at the top of each pagination loop.

EIBAID comparison

In REXX, compare C2X(EIBAID) to a hex string: IF C2X(EIBAID) = 'F3' THEN ... (PF3).
In COBOL, compare EIBAID directly to a hex literal: IF EIBAID = X'F3'